

Joint-Pressure Exact-Pressure Optimization for Simple Zeros of the Riemann Zeta Function

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Abstract

We record a computer-assisted research-draft refinement of the positioned-pressure method for simple zeros of the Riemann zeta function on the critical line. In addition to optimizing a rational 15-term analytic window and exact local pair/position weights, the current construction releases the total local pressure itself as an optimization variable. The resulting seven-point, six-gap inequality is closed by an outward-rounded interval branch-and-bound computation. Subject to the imported analytic interface and finite-dimensional Gram profile of the predecessor lineage, the certified numerical inputs yield

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq 0.67341308528684939167 \dots,$$

i.e. 67.3413085287% before conservative decimal truncation. This manuscript is a research draft, not a peer-reviewed theorem and not a proof of the Riemann hypothesis.

1 Imported analytic interface

Let $S = N_0^s(T, 2T)$ and $N = N(T, 2T)$. We import

$$S \geq HN + \Delta(M) - o(N)$$

and the finite- m Gram profile

$$h_m(E) = \begin{cases} E, & 0 \leq E \leq m/(m-1), \\ E/m + 2\sqrt{(m-1)E/m} - 1, & E \geq m/(m-1). \end{cases}$$

These ingredients belong to the Anthropic / `trmdy` / `sxuff` lineage and are not reproved here.

2 Joint-pressure analytic window

We use

$$v(s) = \sum_{j=0}^{14} c_j \cos(\omega_j s), \quad \omega_0 = \sqrt{2}, \quad \omega_j = 2j\pi \ (1 \leq j \leq 14),$$

with common denominator 10^9 and numerator vector

$$(1000000000, 12948011, -12114181, 3684033, 5911261, -1663892, 5928575, \\ -7167828, 6229914, -5147758, -756341, 440544, -311207, 237969, -190433).$$

Direct interval arithmetic encloses

$$H(v) = 0.672199902667575775421269384482413 \dots$$

and verifies

$$H_{\text{cert}} = 0.6721999026 < H(v).$$

The same interval subdivision proves positivity of the window on the required support.

3 Variable-total-pressure local functional

For six nonnegative gaps let $y_0 = 0$ and $y_j = g_1 + \cdots + g_j$, and set

$$F(g) = \sum_{r=1}^6 b_r g_r + \sum_{0 \leq i < j \leq 6} a_{ij} W(y_j - y_i).$$

The exact nonnegative pair weights are recorded in `candidate.json` and satisfy

$$\sum_{i=0}^{6-s} a_{i,i+s} = 2 \quad (1 \leq s \leq 6).$$

The position-pressure vector is

$$\frac{1}{46000000000}(22420713, 32878293, 37700994, 37700994, 32878293, 22420713),$$

so

$$B = \sum_{r=1}^6 b_r = \frac{93}{23000}.$$

The predecessor block proof uses the position pressure through the symbolic total $B = \sum b_r$, via the summed inequality $E_B + B \text{span}(B) \geq A$. Therefore varying B is admissible provided the new exact total is carried into the global penalty, all pressures remain nonnegative, and the pair-span capacities remain bounded by 2.

4 Six-dimensional interval certificate

Joint adversarial exchange over total pressure, the window, pair weights, and position-pressure distribution found a lowest floating basin near

$$0.0078878193504693 \dots$$

We certify the smaller rational target

$$\varepsilon = \frac{7887}{10^6} = 0.007887.$$

The rigorous verifier uses 4000 interval cells per normalized unit, 50 decimal digits in the outward-rounded table builder, and C++ compiled with `-ffp-contract=off`. It reports

$$\begin{aligned} \text{nodes} &= 3424276, & \text{pruned} &= 1712170, \\ \text{splits} &= 1712106, & \text{convex} &= 1756299, \\ \text{tangent} &= 801918, & \text{max depth} &= 71. \end{aligned}$$

The output is `VERIFIED=true`. Exact table hashes, workflow run 31603343875, artifact 9144421792, and its digest are stored in the repository metadata.

5 Exact shifted-pressure averaging

For an m -point block there are $m - 6$ translated seven-point windows. Set

$$A_m = \varepsilon(m - 6), \quad R_m = h_m(A_m), \quad \eta_m = \frac{R_m}{A_m}.$$

The span-capacity identities control the summed pair contribution by the predecessor block energy, while exact shifted averaging gives pressure charge $(m - 6)B$. Thus

$$\Delta(M) \geq \frac{R_m}{m}S - \frac{\eta_m B(m - 6)}{m}N - o(N),$$

and therefore

$$\boxed{\frac{S}{N} \geq \frac{mH_{\text{cert}} - \eta_m B(m - 6)}{m - R_m} - o(1).}$$

6 Final arithmetic

With

$$H_{\text{cert}} = 0.6721999026, \quad \varepsilon = 0.007887, \quad B = 93/23000,$$

the integer scan selects $m = 145$. Then

$$A_{145} = 1.096293,$$

$$R_{145} = 1.09440740970772755724598196593800412029\dots,$$

and

$$\eta_{145} = 0.99828003071051950276612362382866999998\dots$$

Hence

$$\frac{S}{N} \geq 0.67341308528684939167090457164160\dots - o(1),$$

and safely

$$\boxed{\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} > 0.6734130852.}$$

7 Current tightening experiment

With the exact window and weights fixed, the observed floating minimum still lies about 8.19×10^{-7} above the certified target. A separate hardened run is probing the upward ladder 0.0078875, 0.0078874, 0.0078873, 0.0078872. No value from that ladder enters the statement above unless its interval run closes.

8 Trust boundary and status

The interval window bound, exact pair-span identities, exact pressure total, six-dimensional local certificate, shifted-pressure bookkeeping, and final arithmetic are directly checkable from this repository. The explicit-formula / trace interface and finite- m Gram profile remain imported from prior work. Independent reproduction of the six-dimensional certificate is requested; accordingly, the percentage should be read as an interval-certified research-draft candidate rather than a publication-ready theorem.

Reproducibility

The repository contains exact current parameters in `candidate.json`, verifier evidence in `certificates/`, arithmetic and interval code in `src/`, and superseded records in `archive/`. The repository follows a single-main workflow.